

Assignment 2

Q1:

Let the MDP be defined as $((S, A, P, R, \gamma))$, where:

- (S) : finite set of states
- (A) : finite set of actions
- $(P(s'|s, a))$: transition probability
- $(R(s, a))$: expected immediate reward
- $(\gamma \in (0, 1))$: discount factor

For a deterministic policy π , define:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right]$$

and the state-action value function:

$$Q^\pi(s, a) = R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s')$$

We define a new policy π' based on π as:

$$\pi'(s) = \arg \max_{a \in A} Q^\pi(s, a)$$

that is, in each state, choose the action that maximizes $Q^\pi(s, a)$.

We need to prove that:

$$V^{\pi'}(s) \geq V^\pi(s), \quad \forall s \in S$$

From the Bellman equation for V^π :

$$V^\pi(s) = Q^\pi(s, \pi(s))$$

By the definition of π' :

$$Q^\pi(s, \pi'(s)) \geq Q^\pi(s, \pi(s)) = V^\pi(s)$$

Hence,

$$R(s, \pi'(s)) + \gamma \sum_{s'} P(s'|s, \pi'(s)) V^\pi(s') \geq V^\pi(s)$$

This can be rewritten as:

$$T^{\pi'} V^\pi \geq V^\pi$$

where $T^{\pi'}$ is the Bellman operator for policy π' .

Because $T^{\pi'}$ is a monotone contraction mapping, repeatedly applying it gives:

$$V^{\pi'} = \lim_{k \rightarrow \infty} (T^{\pi'})^k V^\pi \geq V^\pi$$

Thus, every policy improvement step produces a policy that performs no worse than the previous one.

Because the MDP has finite S and A , the total number of possible deterministic policies is finite. Each improvement step satisfies:

$$V^{\pi_{k+1}}(s) \geq V^{\pi_k}(s), \quad \forall s$$

and if $V^{\pi_{k+1}} = V^{\pi_k}$, then $\pi_{k+1} = \pi_k$, i.e. the algorithm is optimal.

Therefore, the sequence of policies

$$\pi_0, \pi_1, \pi_2, \dots$$

is monotonic non-decreasing in value and must converge in a finite number of steps to some policy π^* .

At convergence, π^* satisfies:

$$\pi^*(s) = \arg \max_a Q^{\pi^*}(s, a)$$

which is precisely the optimal policy.

Q2:

We will show that $\{V_n\}$ is a Cauchy sequence.

From the update rule,

$$V_n - V_{n-1} = (1 - \alpha_n)V_{n-1} + \alpha_n x_n - V_{n-1} = \alpha_n (x_n - V_{n-1}).2)$$

Taking absolute values and using boundedness,

$$|V_n - V_{n-1}| = \alpha_n |x_n - V_{n-1}| \leq \alpha_n (|x_n| + |V_{n-1}|) \leq \alpha_n (C_1 + C_2)$$

Because $\alpha_n = 1/n^2$,

$$|V_n - V_{n-1}| \leq \frac{C_1 + C_2}{n^2}$$

For any integers $m > n$,

$$|V_m - V_n| \leq \sum_{k=n+1}^m |V_k - V_{k-1}| \leq (C_1 + C_2) \sum_{k=n+1}^m \frac{1}{k^2}$$

Since $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges, for any $\varepsilon > 0$ there exists N such that whenever $m, n > N$,

$$\sum_{k=n+1}^m \frac{1}{k^2} < \frac{\varepsilon}{C_1 + C_2}$$

Thus,

$$|V_m - V_n| < \varepsilon$$

so $\{V_n\}$ is Cauchy.